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Research Prospectus

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Statistical Foundations of Measures of Association

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Abstract

Across the empirical sciences, deciding whether variables are genuinely related rather than associated by chance or through shared context rests on measures of association. To support principled inference rather than mere description, such measures need a statistical theory describing how they behave, the value it converges to, and how it fluctuates around that value. Classical correlation coefficients come equipped with such a theory, but they are limited to linear or monotonic relationships and to scalar variables, while more flexible alternatives often lack comparable statistical guarantees. My thesis aims at developing flexible measures of association equipped with rigorous statistical methodology. The first part of this work focuses on the Information Imbalance coefficient, a recent rank-based measure whose statistical foundations had been absent. Looking further ahead, my thesis work moves beyond this coefficient to conditional measures of association with comparable guarantees, and their use in causal discovery, fairness, and high-dimensional variable selection.

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1 Introduction

When two variables tend to change together, knowing the value of one tells us something about that of the other. For example, the temperature outside tells us something about how much ice cream a shop will sell; a patient's set of clinical measurements tells us something about their risk of disease. A measure of association turns this intuition into a single number; it quantifies how much knowing one variable reduces our uncertainty about another. In general, the higher the value, the more tightly the two variables are linked. Such measures are foundational tools across statistics and the empirical sciences and they support several distinct tasks. The first one is **variable selection**. When an outcome of interest could depend on a large number of candidate variables, measures of association allow those candidates to be ranked by relevance and the ones that provide negligible information are discarded, often before any model is fitted [15, 30, 16]. The second task is **causal discovery**. While association alone does not establish causation, measures of dependence, particularly those that extend to conditional association, are the basic ingredient of algorithms that reconstruct the network of dependencies among many variables. The third task, and the most common application so far, is **formal hypothesis testing**, i.e., deciding, on the basis of a sample, whether two variables are associated. This testing problem has a precise structure. One begins with a null hypothesis, i.e., the statement that the two variables are independent, and treats the measure of association, computed from the data, as a test statistic. To judge whether the value obtained from sample is significant, one needs its sampling distribution under the null hypothesis, i.e., if the variables were truly unrelated. The observed value is then compared against this reference distribution. If it falls far enough into the tail such that such a value would rarely arise by chance alone, the null hypothesis is rejected and the association is declared significant. The p-value records how extreme the observation was, and the significance level fixes in advance how often one is willing to reject the null when it is in fact true. The entire procedure rests on a single ingredient which is knowledge of how the test statistic behaves under the null. For most measures, this distribution cannot be written down exactly and is instead obtained from asymptotic theory which gives a description of the statistic's behaviour as the sample size grows large.

Herein lies most of the research done towards these measures. The classical measures of association like Pearson's correlation coefficient, Spearman's rank correlation, Kendall's rank correlation coefficient come equipped with exactly this kind of theory, but they are designed for measuring linear or monotonic relationship between pairs of variables. They can miss more intricate forms of association, in some cases even noise-free ones, and are restricted to pairs of single variables; thus, they cannot be used for groups of variables [6]. Real-world data, by contrast, is rarely so simple. We often want to measure association between not just 2 variables but between groups of variables. The newer measures, like constrained covariance, kernel mutual information in [14] and mutual information [19], are built to handle this complexity and are more flexible, but many of them lack statistical guarantees. This prevents their use for hypothesis testing. A measure that is flexible but lacks statistical theory is, in practice, only half a tool.

Information Imbalance (**II**) [12] is a recent distribution-free coefficient of asymmetric association between two groups of variables, built from distance-based ranks within each group. It compares two sets of variables by assessing whether observations that are close to one another in the first space remain close to one another in the second. This allows **II** to handle vector-valued observations directly and to detect association without making assumption on the form of association. It is, in this sense, exactly the kind of flexible measure that modern problems call for. A differentiable version has since been introduced in [29], and the coefficient has been used as a feature selection tool [12] and to detect causality in dynamic processes [8]. What has been lacking is a distributional theory that would allow **II** to be used for principled statistical inference.

The broader aim of my PhD is to develop flexible measures of association that are equipped with statistical guarantees. As a first step, I have worked on supplying this theory for **II**. Establishing its asymptotic limit, i.e., the value it converges to as the sample size grows, gives the coefficient a clear and interpretable meaning, while establishing its asymptotic distribution provides the reference needed to test for association without relying on expensive permutation procedures.

The remainder of this prospectus is organised as follows. Section 2 reviews the existing literature on measures of association, situating **II** among classical and modern alternatives. Section 3 presents the initial work of the PhD, i.e., the asymptotic theory of **II**, comprising its asymptotic limit and its limiting distribution. Section 4 sets out the planned research for the remainder of the PhD, centred on a complete significance-testing framework for **II** and making a contribution towards the broader field of measures of association by developing more flexible coefficients backed with statistical guarantees.

2 Background

Measuring the association between random variables is a foundational challenge in statistics. The relationships between random variables are typically analysed using coefficients of correlation or association, with assessing significance through a test of hypothesis. A common approach to developing such a measure is a rigorous study of its properties: [6], for instance, proposes a set of desirable properties that a measure should ideally satisfy. Once these properties are established, the next task is to develop the asymptotic theory of the measure. With the asymptotic limit and distribution in hand, one can construct a significance testing framework for association, which is one of the primary use cases of any such measure. It is also valuable to analyse the power of a new measure against class of alternative hypothesis, in order to identify the scenarios in which one measure outperforms another.

Classical coefficients such as Pearson's correlation, Spearman's rank correlation, and Kendall's τ are powerful, widely used tools, supported by well-developed asymptotic theories that facilitate hypothesis testing via p-values. These methods are nonetheless fundamentally restricted. They are primarily effective at detecting linear or strictly monotonic associations. When applied to complex data, they often fail to detect non-monotone or highly nonlinear relationships, even in the absence of noise [6].

To address these limitations, statistical research has increasingly shifted toward more flexible methods for testing association. Many such measures have been proposed in the literature: the maximal correlation coefficient [5, 25, 18], coefficients based on joint cumulative distribution functions and ranks [3, 4, 10], kernel-based methods [14, 13, 24], information-theoretic coefficients [22, 19], coefficients based on copulas [9, 23], and coefficients based on pairwise distances [11, 17, 27, 28, 12].

These measures are not without shortcomings. Some lack statistical guarantees, some have asymptotic limits that are not readily interpretable, and others apply only to single variables. The coefficient proposed by [11], for example, has no asymptotic theory, so no analysis of statistical power is possible. By contrast, the measures of [28] and [13] do come with statistical guarantees, and are consequently more widely used. Chatterjee's coefficient [6] has attracted particular attention, largely because of its interpretability; it is an estimator of the measure of association introduced in [9]. However, it acts as a measure of association only when the variables involved are univariate. Subsequent work aimed at greater flexibility led to the [2] coefficient, which detects nonlinear relationships between a single variable and a group of variables. Recently, [12] introduced a rank-based measure of association called Information Imbalance **(II)**, for measuring association between two groups of variables. Its asymptotic theory, however, remains undeveloped.

The theoretical development of Chatterjee's coefficient and its multivariate extension by Azadkia and Chatterjee [2] has been the subject of considerable recent activity, and provides the methodological backdrop for the present work. Because the architecture of **II** closely resembles that of one of these coefficients (asymmetricity, rank-based, and built from nearest-neighbour structure) the techniques developed in this line of work form the natural starting point for its asymptotic analysis. Chatterjee himself established asymptotic normality of his coefficient under the null hypothesis of independence [6, Theorem 2.1]. The corresponding result under general dependence remained open, until [20] resolved it by combining Hájek representations with U-statistic projection arguments to handle the dependencies inherent in nearest-neighbour ranks. The Azadkia-Chatterjee coefficient has attracted a parallel line of work: [7, Theorem 4.1] obtained a central limit theorem under independence in arbitrary metric spaces, [26] derived a closed form for the asymptotic variance, [21] proposed a variant incorporating multiple nearest neighbours to boost its power, and [1] established a central limit theorem under local alternatives.

Drawing on these techniques, the initial part of my PhD develops the corresponding asymptotic theory for **II**. The next section presents this work in detail.

3 Ongoing work

In this section, we first introduce the setup and notation used to define the problem. Next, we describe the results I have established about our problem.

Setup

Let (X,Y) be a random vector taking values in a product metric space $\mathscr{X}\times\mathscr{Y}$, where $\mathscr{X}\subseteq\mathbb{R}^d$ and $\mathscr{Y}\subseteq\mathbb{R}^p$, with joint cumulative distribution function (CDF) $F_{X,Y}$ admitting a continuous density $f_{X,Y}$ with respect to Lebesgue measure, and corresponding continuous marginal densities f_X and f_Y. Let $(X_1,Y_1),\dots,(X_n,Y_n)$ be n i.i.d. copies of (X,Y).

The **(II)** coefficient from X to Y is defined as

$$\widehat{\mathrm{II}}_n(X\to Y) \;=\; \frac{2}{n^2}\sum_{i=1}^{n} R_i^Y(N_X(i)), \tag{1}$$

where $N_X(i)=\arg\min_{k\neq i}\|X_k-X_i\|$ is the index of the nearest neighbour of X_i and

$$R_i^Y(N_X(i)) \;=\; \sum_{j\neq i}\mathbb{1}_{\{\|Y_{N_X(i)}-Y_i\|\,\geq\,\|Y_j-Y_i\|\}}$$

is the rank of $Y_{N_X(i)}$ among the neighbours of Y_i.

The coefficient is *asymmetric*, i.e., $\widehat{\mathrm{II}}_n(X \to Y) \neq \widehat{\mathrm{II}}_n(Y \to X)$ in general, which makes it a directional measure of dependence.

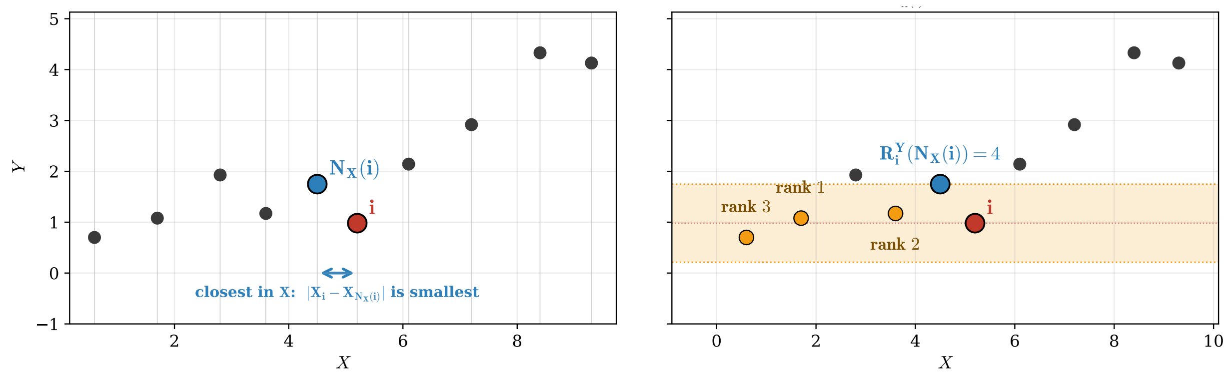

Figure 1. II: rank construction and asymmetry

We now turn to its asymptotic properties.

Asymptotic Limit

As a first step towards developing the asymptotic theory of this coefficient, we worked on finding its asymptotic limit, i.e.; the value this estimator converges to as sample size increases to infinity. The following result gives the limit of $\widehat{\mathrm{II}}_n$:

As $n \to \infty$,

$$\widehat{\mathrm{II}}_n(X \to Y) \xrightarrow{a.s.} \mathrm{II}(X \to Y) := 2P_{Y_1,Y_1',X_1,\tilde{Y}}\left(\|Y_1 - Y_1'\| \geq \|Y_1 - \tilde{Y}\|\right),$$

where $\tilde{Y} \sim f_Y$ is an independent draw from the marginal of Y, and $Y_1, Y_1' \overset{\text{iid}}{\sim} f_{Y|X=X_1}$ are independent copies from the conditional distribution of Y given $X_1 \sim f_X$.

The limit II admits a clear interpretation. It measures the proportion of marginal draws $\tilde{Y}$ that fall within the ball centered at Y_1 with radius $\|Y_1 - Y_1'\|$, where Y_1' is an independent copy having same conditional distribution as that of Y_1.

The authors in [12] conjectured that $\widehat{\mathrm{II}}_n(X \to Y)$ converges to **0** when Y is almost surely a measurable function of X denoting the case of *no imbalance* between variables. It converges to **1** when X and Y are independent of each other denoting the case of *complete imbalance* between variables. We now provide a proof of these results using the limit: in the case of complete independence between X and Y, the limit becomes 1 by exchangeability of $\tilde{Y}$ and Y_1'. In the case when Y is a measurable function of X, the limit evaluates to 0 by the continuity of the conditional density function.

To prove the limit of $\widehat{\mathrm{II}}_n$, we made use of the result: $X_{N_X(i)} \xrightarrow[n\to\infty]{\text{a.s.}} X_i$ (where $N_X(i)$ is the index of nearest neighbour of X_i) ([2, Lemma 11.3]). Sampling a point Y_i from a distribution conditioned on $X_{N_X(i)}$ would be the same as sampling Y_i' from a distribution conditioned on X_i, and hence Y_i, Y_i' can be treated as independent copies drawn from the same distribution under the limit that the sample size is infinite.

To prove the asymptotic limit and asymptotic distribution, we made some assumptions on the densities of variables. The first assumption was that the conditional CDF satisfies the Local Lipschitz condition as mentioned in Assumption A1 of [2]. The second assumption is that the marginal distribution of X has sub-exponential tails as mentioned in Assumption A2 of [2].

Rate of Convergence and Central Limit Theorem

We also found that the rate of convergence of $\widehat{\mathrm{II}}_n$ to its limit is $n^{-\min\{1/2,1/d\}}(\log n)^{d+1+\beta}$, where d is dimension of X and $\beta \geq 0$ is the polynomial growth exponent in the local Lipschitz condition on $f_{Y|X}$ (Assumption A1 of [2]). The rate reflects the rate of convergence of nearest neighbor of X_i to X_i and the rate of convergence of the empirical CDF to the population CDF.

As a next step, we proved the central limit theorem for the coefficient as stated below: For any fixed continuous $f_{X,Y}$ such that Y is not almost surely a measurable function of X,

$$\frac{\widehat{\mathrm{II}}_n - \mathbb{E}[\widehat{\mathrm{II}}_n]}{\sqrt{\mathrm{Var}(\widehat{\mathrm{II}}_n)}} \xrightarrow{d} \mathcal{N}(0,1).$$

This result is a theoretical guarantee that $\widehat{\mathrm{II}}_n$ concentrates around its mean and the fluctuations are asymptotically Gaussian.

The proof utilised results from U-statistics and Hájek projection theory. We further verified all the results using simulations.

4 Research Plan

The field of developing measures of association with well-defined statistical guarantees has been growing rapidly, with new measures being introduced which possess increased flexibility and power. For my PhD, I would like to contribute to this broader field by making theoretical as well as applied contributions.

4.1 Estimation of asymptotic variance

The work done in Section 3 are the first steps towards establishing the asymptotic behaviour of $\widehat{\mathrm{II}}_n$ but leave several important problems open. In this direction, I would like to work on deriving a closed form of the asymptotic variance of the estimator. Our result establishes that the **II** estimator satisfies a central limit theorem of the given form for some asymptotic variance which we denote as σ^2. On its own, this result is not yet usable for hypothesis testing as the variance has no closed-form expression. Characterising σ^2 explicitly is therefore the next primary planned contribution. I further plan to achieve a tighter bound on the rate of decay of bias to yield a CLT centered at population quantity of $\widehat{\mathrm{II}}_n$. Apart from this, I would like to further look into the power of $\widehat{\mathrm{II}}_n$ in measuring association between variables against alternative class of hypothesis.

4.2 Beyond Information Imbalance

A second line of research that I plan to pursue concerns methodological developments for measuring association in more complex data settings. Building on $\widehat{\mathrm{II}}_n$, I would like to develop new coefficients capable of quantifying associations in multivariate spaces, extending beyond pairwise relationships. I further plan to derive **conditional versions** of these coefficients to measure conditional dependence, which isolates the direct association between two variables from associations mediated by other variables. As with the work on $\widehat{\mathrm{II}}_n$, the theoretical investigation of these measures will include the study of their consistency, asymptotic behaviour, and other statistical properties. Apart from this, I would like to explore analogous frameworks in settings where variables are discretely distributed, so that the methodology applies to both continuous and discrete data.

4.3 Applications

Alongside these theoretical developments, I plan to study the proposed methodology in several modern applications in statistics and machine learning.

A first application is **causal discovery**. Recovering causal structure from observational data is notoriously difficult: classical correlation-based measures are symmetric and cannot distinguish cause from effect, and observed dependences may be induced by unobserved common causes rather than direct causal links. A coefficient that is both conditional and directional addresses both difficulties at once. Conditioning screens out spurious dependences, while asymmetry provides a signal for orienting relationships. I would like to investigate how the coefficients developed in this project can contribute to this area, both as building blocks for testing individual causal hypotheses and, more ambitiously, as ingredients in algorithms for learning causal structure from data. A second application is **fairness** in machine learning. As predictive models are increasingly deployed in consequential settings such as hiring, lending, and healthcare, there is growing demand for principled tools to assess whether predictions depend on sensitive attributes, either directly or through proxies. Many fairness criteria can be expressed as statements about dependence, or conditional independence, between predictions, outcomes, and sensitive variables, making flexible measures of association a natural tool. They can serve both as diagnostics and as building blocks for fairness-aware training procedures that penalise unwanted dependence. I plan to explore both roles, with particular attention to the finite-sample guarantees needed for fairness audits to be statistically meaningful. A third application is consistent **variable selection** in high-dimensional settings. When the number of candidate variables exceeds the number of observations, identifying a parsimonious subset that genuinely drives the response is essential for interpretability and downstream inference. Selection procedures built on measures of association are attractive because they are nonparametric and not tied to a specific model class, but they require measures that remain well-behaved in high dimensions and admit consistency-type selection guarantees. I would like to investigate the extent to which the coefficients developed in this project meet these requirements, and how they can be used to construct selection procedures with provable guarantees.

Taken together, these applications form a coherent programme in which the project's theoretical and methodological developments are stress-tested through contact with substantive statistical and machine learning problems.

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